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Course 608

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Research proposal

Incorporation and analysis of partially carbonated calcium hydroxide particles for the improvement in industrial applicability of Limestone Calcined Clay Cement.

Ernest Goh

Grade 11

NUS High School of Mathematics and Science Research proposal

Singapore

Submitted on 20/9//2026

Rationale

Concrete is the second most consumed material on earth, after water. However, the production of cement, a significant component of concrete, also produces significant amounts of carbon dioxide emissions. The production of 1 tonne of Ordinary Portland Cement (OPC), releases roughly 0.85 tonnes of CO_2 emissions per tonne manufactured. In total, this accounts for 8% of global CO_2 emissions.[1] As cement demand is expected to keep pace with global construction growth, reducing the carbon emissions of cement is one of the most consequential actions available to decarbonising the built environment. Hence, researchers have been actively looking to substitute the use of cement in concrete with other materials. One of the most widely explored and most promising OPC alternatives is Limestone Calcined Clay Cement (LC3), which replaces up to 50% of OPC with a mixture of limestone ($CaCO_3$) and calcined clay, lowering the carbon emissions of cement by up to 40%.[2] These raw materials are far more abundant than the fly ash or slag used in other supplementary cementitious materials. This combination of performance, availability and compatibility with existing clinker production infrastructure has made LC3 a leading candidate for near-term, large-scale decarbonisation of the cement industry.

However, LC3 has certain trade-offs when compared to OPC. LC3 exhibits lower early age strength as compared to OPC. This is due to the pozzolanic reaction between calcined clay and calcium hydroxide contributing little strength gain in the first hours of hydration, as well as the lowered OPC content in LC3.[3] This early strength deficit is repeatedly identified in literature as the single largest barrier to LC3's commercial uptake, particularly in fast paced construction projects that require high early strength. It has also been observed that beyond 28 days, there is little improvement in strength, while OPC continues to become stronger.[2] This could be due to a combination of multiple factors, including the scarcity of reactive calcium hydroxide, and a more refined pore structure. In traditional OPC systems, the calcium hydroxide generated by the reaction acts as a sacrificial buffer to maintain the high pH of the system to protect the steel rebar used.[4] The calcium hydroxide in traditional OPC systems also acts to help heal inevitable microcracks in concrete.[5] However, due to the diminished calcium hydroxide content in LC3 systems, such benefits are often not seen in LC3 systems.

Calcium carbonate, the primary component of limestone, can be formed through the carbonation reaction of calcium hydroxide and carbon dioxide. When calcium hydroxide is carbonated in air, the carbonation reaction tends to spread from the outside inwards. This results in the formation of a protective outer calcium carbonate layer around the calcium hydroxide particle. [6]

Research question

This research aims to address the various trade-offs of LC3 in order to increase its industrial applicability to become a better OPC alternative. Literature has shown that the surface of the carbonation shell formed during carbonation provides high energy nucleation sites for the formation of hydration products, thereby accelerating early age strength development.[7] Additionally, the addition of CH has shown to address the late-stage plateau in compressive strength by promoting the pozzolanic reaction of metakaolin.[3] However, the production of calcium hydroxide undermines the environmental advantage of LC3

As such, this project proposes to combine both solutions into one, by partially carbonating calcium hydroxide particles and applying it to LC3 systems. The partially carbonated shell

would be able to promote early-stage strength development by providing high energy nucleation sites while also allowing for the addition of calcium hydroxide to address late-stage strength and other trade-offs, as well as reducing the carbon footprint of calcium hydroxide through the uptake of carbon dioxide during the carbonation reaction. The project will produce and characterize the shell-core structure of partially carbonated calcium hydroxide particles and investigate whether the addition will improve the industrial applicability of LC3 systems. This will be done by varying the particle size of calcium hydroxide particles, and the carbonation duration.

Hypothesis

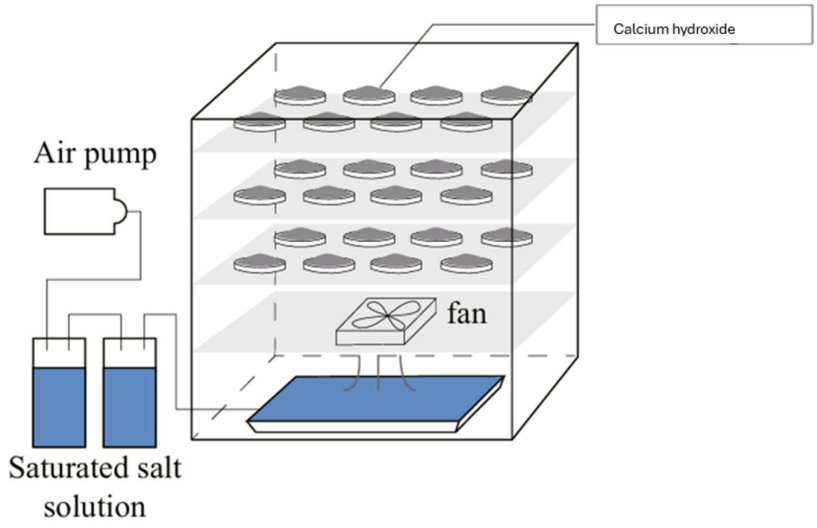
I hypothesize that the addition of smaller lime particles would result in a larger shell to core volume ratio due to their higher surface area to volume ratio. I also hypothesize that the addition of partially carbonated lime particles will exhibit improved early age compressive strength relative to LC3 containing ground limestone, due to the presence of high energy nucleation sites on the carbonate shell. The partially carbonated lime particles will exhibit improved late-stage compressive strength as the addition of reactive calcium hydroxide will help to sustain the pozzolanic reaction of calcined clay.

Expected outcomes

It is expected that the progressive carbonation of lime particles will result in the formation of a measurable carbonate shell-like outer region, with the thickness and characteristics being influenced by the duration carbonated and the size of the particle. It is also expected that the addition of partially carbonated lime particles would help the mechanical performance of the LC3 system, exhibiting better early stage compressive strength and potentially better late stage compressive strength as well.

Methodology

Carbonation of calcium hydroxide particles



1. A chamber will be modified to house the lime particle samples. The chamber is pictured above.
    a. The outlet of the chamber will be left open, and the inlet will be connected to bottles containing a saturated salt solution of K_2SO_4 and an air pump. Air would be bubbled through the saturated salt solution and then fed into the chamber.
    b. In the chamber, a plate containing the saturated salt solution is placed at the bottom, alongside a fan. The fan would help circulate air, and the saturated salt solutions is used to maintain a constant humidity of roughly 95%. This is used to accelerate the carbonation reaction.
    c. A humidity meter will be placed inside the chamber to ensure the humidity of the chamber remains constant.
2. A thin layer of calcium hydroxide particles is placed on a plastic dish, ensuring the layer is as thin as possible to ensure all particles are in contact with the air. This is then repeated multiple times in multiple plates.
3. The plates are then placed in a chamber for different durations.

Incorporation of partially carbonated lime in LC3

1. Once removed from the chamber, the partially carbonated lime particles are then mixed into a mixture containing OPC, gypsum, calcined clay, limestone and water. The mixing ratio will follow the ratio for LC3-50. The limestone component would be substituted by the partially carbonated lime progressively in intervals of 25%.
2. The mixture is then poured into cube moulds and left to cure.
3. At set intervals, the cubes would be crushed by a universal testing machine to determine the compressive strength of the cube.

Data analysis

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After partial carbonation, lime particles will be placed in an X-Ray Diffraction (XRD) machine. This will be used to determine the degree of carbonation, which will indicate how much calcium carbonate has been formed in the reaction. It will also give a breakdown of the different phases of carbonate formed. Some carbonate phases, like vaterite, are more reactive than others, like calcite, which could have added benefits to the LC3 system.

Lime particles will also be analysed with a scanning electron microscope (SEM) using a technique called backscattering electrons (BSE) image analysis, where the sample would be immersed in epoxy and polished, opening the calcium hydroxide particles. This would allow us to see the carbonate shell formed and analyse its thickness and other characteristics.

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A Universal Testing Machine (UTM) will be used to crush the cubes of concrete formed. This would allow us to determine the compressive strength of the concrete cubes formed.

XRD will also be used to determine the amount of CH within the system as time progresses.

SEM will also be used to determine the structure of the concrete formed.

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